Evidence first.
Curiosity always.

A connected portfolio of reproducible neuroscience experiments, applied neurotechnology, and communication artifacts.

PROJECT	VERIFIED STATUS	PRIMARY EVIDENCE
Mouse visual cortex	One-mouse analysis complete; replication pending	Sensorium report, six notebooks, figures
EEG eye state	First time-aware study complete	Python analysis, metrics, report, explainer
Neurodivergent / Neurocomms	Applied project references and templates collected	Mobile, HTML, presentation, PDF, video patterns

1 / Portfolio thesis

We study how neural activity relates to behavior, then translate the work into accessible tools and explanations. The portfolio moves from a scientific question to a public, reviewable artifact without hiding uncertainty or collaboration boundaries.

2 / Flagship study

The mouse visual-cortex project compares stimulus-only and behavior-augmented ridge-regression encoding models on Sensorium recordings. Under the same pipeline, the original mouse rises from 0.1265 to 0.1583 and a second mouse rises from 0.1543 to 0.2034. Pupil improves both animals; running contributes more in the second. The claim remains about prediction, not causation.

3 / Complementary study

The EEG eye-state project tests whether causal temporal context improves classification from a public 14-channel EEG recording. The pointwise baseline reaches test ROC AUC 0.625; the first context extension reaches 0.530 on the final chronological block. The negative result is retained because validation improvement did not guarantee held-out generalization.

4 / Translation layer

The neuro-divergent and neurocomms work contributes human-centered patterns for communication and accessibility applications. Those patterns are reused as mobile specifications, HTML explainers, presentation surfaces, PDF examples, video scripts, and scope/limitations notes for the research projects.

5 / Evidence and reproducibility

The umbrella repository contains the reviewer index, contribution ledger, reproducibility checklist, interactive explainers, project reports, presentation outlines, and video production gates. The flagship repository now includes a core dependency manifest and a written replication protocol. The EEG project retains the downloaded ARFF data, provenance note, analysis script, metrics JSON, and report.

6 / Honest remaining work

Before submission, the applicant must personalize the essays with genuine experiences, run the mouse replication protocol across additional animals, and record the final videos in the applicant’s own voice. This packet does not present those items as complete.

Portfolio links

GitHub hub: github.com/mehtama1234/ASDRP-Neuroscience-Portfolio
Flagship code: github.com/AmishI123/neuroscience-project
Live portfolio: asdrp-neuroscience-portfolio.gargibhatia37240.chatgpt.site

Generated from the committed application materials. Source Markdown remains alongside this PDF.